A4 for SDA (data) and A5 for SCL (clock). These two pins can be connected to multiple devices on a single bus, where all the SDA pins share a common connection, as do all of the SCL pins.

For the following project, we will use the DS1307 RTC connected to the Arduino's I2C bus to display the current date and time. Because we might not want Mountain Standard Time (if you bought the version from SparkFun), we will also provide a sketch for setting the current date and time. This is also useful because, while the RTC is extremely accurate, it has a tendency to shift about ±1 minute every month. The sketches provided try to keep things as simple as possible, leaving the creative application up to the reader.

Hooking It Up

The following schematic and illustration in Figures 10-5 and 10-6 will make the assumption that the RTC is the only device connected to the I2C bus. If we needed to add additional I2C devices, we would need to also add a pair of 10 kilohm pull-up resistors connecting each line of the I2C bus to the positive supply. Otherwise, the wiring is fairly straightforward with a total of four wires needed to get our clock up and running.

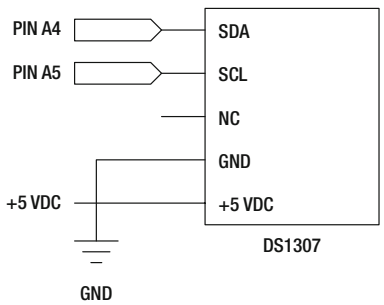

Figure 10-5. *Serial Time Clock schematic*

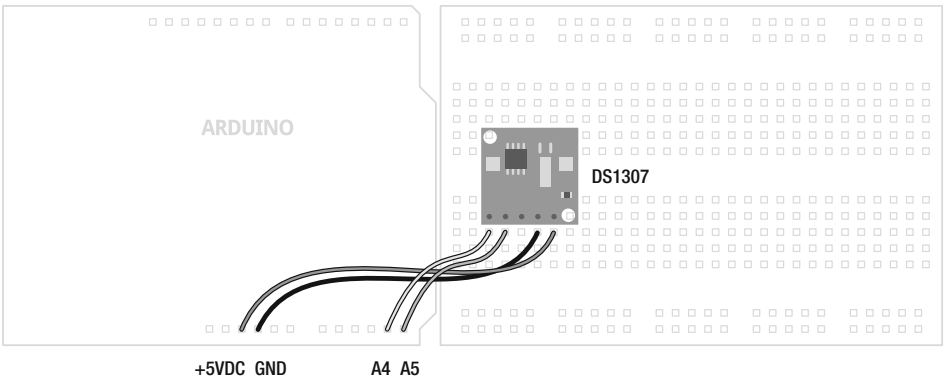

Figure 10-6. *Serial Time Clock illustration*